

SC7 Bayes Methods MT25

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Lecture 11 : MCMC updates using transformations
(preamble to model averaging with Reversible-Jump MCMC)

MCMC using transformations in fixed dimension

Target $\pi(d\theta) = \pi(\theta)d\theta$ with $\theta \in \Omega$.

Take density $g(u)$, $u \in \mathcal{U}$ and function $\psi_1(\theta, u)$ invertible for u at fixed θ . Given θ , set $\theta' = \psi_1(\theta, u)$ with $u \sim g(u)$. Then

$$q(d\theta'|\theta) = g(u)du \quad (1)$$

gives the change of variables under an integral so the density is

$$q(\theta'|\theta) = g(u) \left| \frac{\partial \theta'}{\partial u} \right|^{-1}$$

Example: for $a > 0$ set $\theta' = \theta + a(2u - 1)$ with $u \sim U(0, 1)$,

$$g(u) = \mathbb{I}_{0 < u < 1}, \quad \psi_1(\theta, u) = \theta + a(2u - 1), \quad q(\theta'|\theta) = \frac{\mathbb{I}_{\theta - a < \theta' < \theta + a}}{2a}.$$

If the value $u' \in \mathcal{U}$ reverses the move, so $\psi_1(\theta', u') = \theta$, then

$$\psi_1(\psi_1(\theta, u), u') = \theta. \quad (2)$$

Example: In the example above, check (2) for $u' = 1 - u$.

Let $u' = \psi_2(\theta, u)$ be the function solving (2). We map pairs

$$(\theta', u') = (\psi_1(\theta, u), \psi_2(\theta, u)).$$

If $\psi = (\psi_1, \psi_2)$ then ψ is an *involution*,

$$(\theta, u) = \psi(\psi(\theta, u)).$$

Example: if $\psi(\theta, u) = (\theta + a(2u - 1), 1 - u)$ as before then

$$\begin{aligned}\psi(\psi(\theta, u)) &= \psi(\theta + a(2u - 1), (1 - u)) \\ &= (\theta + a(2u - 1) + a(2(1 - u) - 1), 1 - (1 - u)) \\ &= (\theta, u)\end{aligned}$$

Up till now we chose $q(\theta'|\theta)$ and found a density $g(u)$ and a function $\theta' = \psi_1(\theta, u)$ to simulate it. Let's just write down g and ψ_1 and let q be whatever it is. This gives more freedom.

Theorem: let $\psi : \Omega \times \mathcal{U} \rightarrow \Omega \times \mathcal{U}$ be an invertible, differentiable involution with $\theta, \theta' \in \Omega$ and $u, u' \in \mathcal{U}$.

The MCMC update with proposal $u \sim g(u)$, $(\theta', u') = \psi(\theta, u)$, acceptance ratio

$$r(\theta', u' | \theta, u) = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u), \quad (3)$$

with J_ψ the Jacobian for the transformation $(\theta', u') = \psi(\theta, u)$,

$$J_\psi(\theta, u) = \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right|,$$

and acceptance probability

$$\alpha(\theta' | \theta) = \min \left\{ 1, r(\theta', u' | \theta, u) \right\} \quad (4)$$

satisfies DB in Eqn 7 with q given in Eqn 1.

Proof: Let

$$K(\theta, d\theta') = \alpha(\theta'|\theta)q(d\theta'|\theta) + c(\theta)\delta_\theta(d\theta') \quad (5)$$

with c the probability to reject,

$$c(\theta) = \left[1 - \int_{\Omega} q(d\theta'|\theta)\alpha(\theta'|\theta) \right].$$

K is the transition kernel K for the MCMC. DB is

$$\int_B \pi(d\theta') \int_A K(\theta', d\theta) = \int_A \pi(d\theta) \int_B K(\theta, d\theta') \quad (6)$$

for all measurable A, B . If this holds the process is stationary,

$$\pi(B) = \int_{\Omega} \pi(d\theta) K(\theta, B).$$

Sub (5) into (6). The $c(\theta)$ -term cancels, so DB is

$$\pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta') = \pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta). \quad (7)$$

Check for $\theta' \neq \theta$.

Verify (7) for g , ψ and α in the theorem.

$$\pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta') = \pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta) \quad (8)$$

Change variables in (8) from θ' to u at fixed θ using Eqn 1,

$$\pi(d\theta)q(d\theta'|\theta)\alpha(\theta'|\theta) = \pi(\theta)g(u)\alpha(\theta'(\theta, u)|\theta)dud\theta.$$

If $r(\theta', u'|\theta, u) = 1/r(\theta, u|\theta', u')$ then $r(\theta', u'|\theta, u) \leq 1$ WLOG.

$$RHS = \pi(\theta)g(u)\frac{\pi(\theta')g(u')}{\pi(\theta)g(u)}J_{\psi}(\theta, u)dud\theta$$

$$= \pi(\theta')g(u')\left|\frac{\partial(\theta', u')}{\partial(\theta, u)}\right|dud\theta$$

$$= \pi(\theta')g(u')du'd\theta'$$

$$= \pi(d\theta')q(d\theta|\theta')\alpha(\theta|\theta')$$

making a change of variables and using $\alpha(\theta|\theta') = 1$.

We assumed

$$r(\theta', u' | \theta, u) = 1/r(\theta, u | \theta', u'), \quad (9)$$

so we could take $r(\theta', u' | \theta, u) \leq 1$ **WLOG**. Verify this. From (3)

$$1/r(\theta, u | \theta', u') = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta', u')^{-1}.$$

and

$$r(\theta', u' | \theta, u) = \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} J_\psi(\theta, u)$$

so we need $J_\psi(\theta', u')^{-1} \Big|_{(\theta', u') = \psi(\theta, u)} = J_\psi(\theta, u)$. We know

$$J_\psi(\theta', u')^{-1} \Big|_{(\theta', u') = \psi(\theta, u)} = J_{\psi^{-1}}(\theta, u).$$

Now $(\theta, u) = \psi(\psi(\theta, u))$ is useful: ψ is ψ^{-1} .

□

The Jacobian must be non-singular, so that the change of variables is well defined.

We certainly need $\dim(\theta', u') = \dim(\theta, u)$. This condition is called “dimension matching”.

Example: “Random walk on a log scale”. Suppose we are targeting $\theta \sim \exp(1)$ and we use the proposal

$u \sim U(1/2, 2)$, $\theta' = u\theta$ so that $(\theta', u') = (u\theta, 1/u)$
 $g(u) = 1/(2 - 0.5)$, $0.5 < u < 2$ and $\dim(\theta', u') = \dim(\theta, u) = 2$
so dimensions match. The Jacobian is $1/u$ since

$$\left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| = \begin{vmatrix} u & \theta \\ 0 & -1/u^2 \end{vmatrix} = 1/u.$$

The algorithm is as follows. If $X_t = \theta$ then

- 1) simulate $u \sim U(1/2, 2)$ and set $\theta' = u\theta$;
- 2) with probability

$$\alpha(\theta'|\theta) = \min \left\{ 1, \frac{\pi(\theta')g(u')}{\pi(\theta)g(u)} \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| \right\} = \min \left\{ 1, e^{-\theta' + \theta} u^{-1} \right\},$$

set $X_{t+1} = \theta'$ and otherwise $X_{t+1} = \theta$.

Factors of $g(u)/g(u')$ cancel in α . This proposal is useful if simulating a density which is peaked or diverges at a boundary. This whole framework is a useful restatement of MCMC.

Exercise: show that the Jacobian for the SRW proposal

$$u \sim U(-a, a), \quad (\theta', u') = (\theta + u, -u)$$

is equal one.

Exercise: in the `swiss`-data regression example in lecture 10, I gave the MCMC updates for z and θ . The update for σ is

$$u \sim U(\delta, 1/\delta), \quad \sigma' = u\sigma$$

with $0 < \delta < 1$ a constant we can adjust for efficient MCMC. Calculate the acceptance probability (answer in R-code).

Matched proposals

When we have a mixture of N kernels, detailed balance is

$$\sum_{i=1}^N \pi(d\theta) \xi_i K_i(\theta, d\theta') = \sum_{j=1}^N \pi(d\theta') \xi_j K_j(\theta', d\theta).$$

and up to now this held because

$$\pi(d\theta) \xi_i K_i(\theta, d\theta') = \pi(d\theta') \xi_i K_i(\theta', d\theta), \quad i = 1, \dots, N$$

as each kernel satisfied DB by itself. This isn't necessary.

Let σ_i be the index of the kernel paired with K_i . If kernel i goes with $j = \sigma_i$ then K_j must pair with $i = \sigma_j$. It is enough to have

$$\pi(d\theta) \xi_i K_i(\theta, d\theta') = \pi(d\theta') \xi_{\sigma_i} K_{\sigma_i}(\theta', d\theta), \quad i = 1, \dots, N.$$

This works for

$$K_i(\theta, d\theta') = q_i(d\theta'|\theta)\alpha_i(\theta'|\theta) + c_i(\theta)\delta_\theta(d\theta')$$

with $q_i(d\theta'|\theta) = g_i(u)du$ and $j = \sigma_i$ if

$$\begin{aligned}\alpha_i(\theta'|\theta) &= \min \left\{ 1, \frac{\pi(\theta') \xi_j q_j(\theta|\theta')}{\pi(\theta) \xi_i q_i(\theta'|\theta)} \right\} \\ &= \min \left\{ 1, \frac{\pi(\theta') \xi_{\sigma_i} g_{\sigma_i}(u')}{\pi(\theta) \xi_i g_i(u)} \left| \frac{\partial(\theta', u')}{\partial(\theta, u)} \right| \right\}. \quad (10)\end{aligned}$$

When we do this *individual updates* won't be reversible. We don't even need $q_i(\theta'|\theta) > 0 \Rightarrow q_i(\theta|\theta') > 0$. However the chain as a whole is still reversible.

It is sufficient to check (10) satisfies

$$\pi(d\theta)\xi_i q_i(d\theta'|\theta)\alpha_i(\theta'|\theta) = \pi(d\theta')\xi_j q_j(d\theta|\theta')\alpha_j(\theta|\theta').$$

The proof is essentially the same as before.

Eg: $\theta \sim \exp(1)$ with $u \sim U(1, 2)$ wp $1/2$ else $u \sim U(0.5, 1)$

The mapping $(\theta', u') = (u\theta, 1/u)$ and Jacobian are unchanged.

The algorithm is

(1) with probability $1/2$

(a) set $u \sim U(1, 2)$ and otherwise (b) set $u \sim U(0.5, 1)$.

(2) Set $\theta' = u\theta$.

(3) if we chose (a) then

$$g_2(u')/g_1(u) = \frac{U(u'; 0.5, 1)}{U(u; 1, 2)} = 2.$$

Since $\rho_1 = \rho_2 = 1/2$, we accept θ' with probability

$$\alpha(\theta'|\theta) = \min\{1, 2 \times e^{-\theta' + \theta - \log(u)}\},$$

and if we chose (b) then $g_2(u')/g_1(u) = 0.5$ and accept θ' wp

$$\alpha(\theta'|\theta) = \min\{1, 0.5 \times e^{-\theta' + \theta - \log(u)}\}.$$